NAME : .....

ALLEN CAREER INSTITUTE KOTA (RAJASTHAN) — path to success

RACE
Regular Analysis through Continuous Exercise

JEE (Main + Advanced) 2023
ENTHUSIAST + LEADER COURSE

SCORE ADVANCED | **MATHEMATICS** | **M.M. 76**

REVISION ASSINGMENT # 08 | (CONIC SECTION-02) | TIME : 70 MIN.

SECTION–I(i)

Straight Objective Type (3 Marks each, –1 for wrong answer)

1. If latus rectum of hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ subtends right angle at point (0,b), then square of eccentricity of the hyperbola is equal to -

(A) $2-\sqrt{2}$ (B) 2 (C) $2+\sqrt{2}$ (D) $2\sqrt{2}$

2. The slope of a chord of the parabola $y^2 = 4ax$ which is normal at one end and which subtends a right angle at the origin is-

(A) $\frac{1}{\sqrt{2}}$ (B) $\sqrt{2}$ (C) 2 (D) $\frac{1}{2}$

3. Let a line $\ell x - y - 2\ell = 0$ (where $\ell \in R - \{0\}$) intersects the parabola $y^2 = 8x$ at points A and B. Tangents drawn to parabola at A and B intersect at C(–2, 5), then sum of the angles $\angle CAB$ and $\angle CBA$ is -

(A) π (B) $\frac{\pi}{2}$ (C) $\frac{\pi}{3}$ (D) $\frac{2\pi}{3}$

4. The number of values of c such that the line $y = cx + c$, $c \in R$ touches the curve $\frac{x^2}{4}+\frac{y^2}{1}=1$ is

(A) 0 (B) 1 (C) 2 (D) more than 2

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5. Let $\frac{x}{2}\cos\theta+\frac{y}{1}\sin\theta=1$, $\theta\in\left(\frac{\pi}{18},\frac{\pi}{15}\right)$ intersect the ellipse $x^2 + 2y^2 = 6$ at P and Q, then angle between tangents at P and Q of the ellipse is -

(A) $\frac{\pi}{16}$ (B) $\frac{\pi}{6}$ (C) $\frac{\pi}{8}$ (D) $\frac{\pi}{2}$

6. If mid point of the chord intercepted on the line $9x - 4y = 13$ by the ellipse $\frac{x^2}{4}+\frac{y^2}{9}=1$ is (α, β), then $(\alpha + \beta)$ is

(A) 0 (B) 2 (C) 1 (D) –2

SECTION–I(ii)

Multiple Correct Answer Type (4 Marks each, –1 for wrong answer)

7. Equation of a minor axis of an ellipse is $x + 2y = 3$ and length of minor axis is $\sqrt{5}$, foot of directrix is (2, 3), then

(A) equation of ellipse is $4(2x - y - 1)^2 + 2(x + 2y - 3)^2 = 25$

(B) eccentricity of ellipse is $\frac{1}{\sqrt{2}}$

(C) major axis is $\sqrt{10}$

(D) length of LR is $\frac{\sqrt{5}}{\sqrt{2}}$

8. Ray $y - 2 = 0$ strike the parabola $y^2 = 4x$ at A and after reflection it passes through a point B on the axis of parabola & normal at point A meet the axis of parabola at C, then co-ordinates of -

(A) $A \equiv (2, 0)$ (B) $B \equiv (1, 0)$ (C) $B \equiv (3, -6)$ (D) $C \equiv (3, 0)$

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9. From a point on $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$, two tangents are drawn to $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ and the corresponding chord of contact meets $\frac{x^2}{a^2}-\frac{y^2}{b^2}=0$ at points Q and R then

(A) The locus of mid-point of QR is $x^2 + y^2 = a^2$

(B) The locus of mid-point of QR is $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$

(C) The locus of image of mid-point of QR in x-axis is $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$

(D) The locus of foot of perpendicular from point $\left(\sqrt{a^2+b^2},0\right)$ to QR is $x^2 + y^2 = a^2$

10. If the eccentricity of the ellipse $\frac{x^2}{(\log a)^2}+\frac{y^2}{(\log b)^2}=1$ $(a>b>0, a, b\neq 1)$ is $\frac{1}{\sqrt{2}}$ and 'e' be the eccentricity of the hyperbola $\frac{x^2}{(\log_b a)^2}-y^2=1$, then e^2 is greater than (where logx = ℓnx)-

(A) $\frac{3}{2}$ (B) $\frac{1}{2}$ (C) $\frac{2}{3}$ (D) $\frac{5}{4}$

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11. Consider a curve $C: y=\sqrt{\frac{3x^2-21}{7}}$ and a line L of slope 1, which is tangent to C at point A, then

(A) Equation of L is $y = x + 2$
(B) Equation of L is $y = x - 2$
(C) Equation of circle passing through origin and touches the curve C at A is $x^2 + y^2 + x - 41y = 0$
(D) Equation of circle passing through origin and touches the curve C at A is $4x^2 + 4y^2 + x - 41y = 0$

12. If the tangent at the point P having abscissa $\frac{\sqrt{325}}{2}$ to the curve $\frac{x^2}{25}-\frac{y^2}{16}=1$ meets the director circle of the circle $2x^2 + 2y^2 = 25$ at (α, β) & (γ, δ), then numerical value of harmonic mean of β and δ is greater than

(A) 7 (B) 6 (C) 5 (D) 4

Space for Rough Work

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SECTION–I(iii)

Linked Comprehension Type (Single Correct Answer Type) (3 Marks each, –1 for wrong answer)

Paragraph for Question 13 and 14

Let H : $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ is a hyperbola such that its eccentricity and eccentricity of its conjugate hyperbola are same. Let foot of the perpendicular drawn from focus of H to its one of the tangent is $\left(\frac{1}{2},\frac{\sqrt{15}}{2}\right)$.

Tangents drawn at the ends of chord of length 2 unit of the circle C : $x^2 + y^2 = a^3 - 4b^2 + 12$, meets each other at 'P' on the transverse axis of H.

On the basis of above information, answer the following questions :

13. Number of common tangent of H & C is

(A) 2 (B) 4 (C) 6 (D) 8

14. Let E : $\frac{x^2}{16}+\frac{y^2}{\beta^2}=1$ be the ellipse which cuts H orthogonally then β is

(A) $4\sqrt{2}$ (B) $3\sqrt{2}$ (C) $2\sqrt{2}$ (D) $\sqrt{2}$

Paragraph for Question 15 and 16

Let x_1, x_2, x_3, x_4 be non zero real numbers. Let A,B,C,D be distinct points on the parabola $y^2 = 4x$. Suppose that $F(x_0,y_0)$ is mid point of focal chord AB and lines BC and AE are parallel lines, where $A(x_1,y_1)$, $B(x_2,y_2)$, $C(x_3,y_3)$, $D(x_4,y_4)$ and E(2,0).

15. The value of y_3 is-

(A) $4y_0$ (B) y_0 (C) $\frac{y_0}{2}$ (D) $2y_0$

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16. If $y_1y_4 = 4$, then the tangent at D and normal at A to the parabola meet at a point whose abscissa is-

(A) $\frac{1-x_0}{2}$ (B) $1 + x_0 - x_2$ (C) $1 - x_0 + x_2$ (D) $\frac{1+x_0}{2}$

SECTION–I(v)

Matching list type (4 × 4) (Single option correct) (3 Marks each, –1 for wrong answer)

17. A hyperbola 'H' intersects the ellipse $\frac{x^2}{9}+\frac{y^2}{5}=1$ orthogonally. The eccentricity of the hyperbola 'H' is three times the eccentricity of the ellipse. Transverse and conjugate axis of H are along x and y axis respectively. If 2a, 2b are respectively the length of transverse axis and conjugate axis, $(x_1, 0)$ is the focus and ℓ is length of latus rectum of H.

Match List-I with List-II and select the correct answer using the code given below the list.

	List-I		**List-II**
(P)	a	(1)	1
(Q)	$\lvert x_1\rvert$	(2)	2
(R)	b^2	(3)	3
(S)	ℓ	(4)	6

Codes :

	P	Q	R	S
(A)	1	2	3	4
(B)	1	2	4	3
(C)	4	3	2	1
(D)	1	4	3	2

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18. Let $\ell x + my + 1 = 0$ is common tangent of curves $x^2 = 4y$, $x^2 + \frac{y^2}{3} = \frac{1}{4}$ and $y^2 = 4ax$, $a > 0$.

Match List-I with List-II and select the correct answer using the code given below the list.

List-I		**List-II**	
(P)	$4\ell^2 - a$	(1)	neither prime nor composite number
(Q)	$3m^2 + 3a$	(2)	odd composite number
(R)	$\frac{5\ell - a}{4m}$	(3)	odd prime number
(S)	$2a^2 + 3\ell^2 + 4m$	(4)	even composite number

Codes :

	P	Q	R	S
(A)	1	2	4	3
(B)	1	4	2	3
(C)	2	3	4	1
(D)	3	4	1	2

SECTION–II (iii)

Numerical Grid Type (Upto Second Decimal place) (4 Marks each, –1 for wrong answer)

1. Area of triangle formed by common tangents to parabola $y^2 = 8x$ and hyperbola $\frac{x^2}{1} - \frac{y^2}{3} = 1$ and directrix of parabola is

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2. A parabola passes through two points A(1,0) and B(–1,0) such that directrix always touches the $x^2 + y^2 = 4$ and if locus of focus of parabola is $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, then $\frac{a^2}{b^2}$ is

3. Let A be any point on the ellipse $x^2 + 16y^2 = 4$ with foci S_1 and S_2 then the maximum value of $AS_1 . AS_2$ is

4. From any point on line $(t + 2)(x + y) = 1$, $t \neq -2$ tangents are drawn to ellipse $4x^2 + 16y^2 = 1$. If chord of contact passes through a fixed point, then number of integral values of t for which fixed point lies inside ellipse is

REVISION ASSINGMENT # 07 (CONIC SECTION-01) MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	D	C	B	D	A	C	A	B	D	D
	Q.	11	12	13	14	15	16	17	18	19	20
	A.	B,D	A,D	A,C	A,B,C	A,B,C,D	A,C	B	D	D	A
SECTION-III	Q.	1	2	3	4	5					
	A.	2	1	9	1	5					

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